Quantified asymptotic analysis for the relativistic quantum mechanical system with electromagnetic fields

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Shih-Hsien Fest on Hyperbolic Conservation Laws and Kinetic Theory

April 5, 2024

Introduction

Relating physical models in different scalings

- A Cucker-Smale flocking model:

$$\frac{\mathrm{d}x_i}{\mathrm{d}t} = v_i, \quad \frac{\mathrm{d}v_i}{\mathrm{d}t} = \frac{1}{N}\sum_{k=1}^{N} \phi(x_i - x_k)(v_k - v_i).$$

- When $N \gg 1$, the dynamics of entire ensemble is approximated by kinetic equation:

$$\partial_t f + v \cdot \nabla_x f + \nabla_v \cdot (F[f]f) = 0,$$
$$F[f](t,x,v) := \int_{\mathbb{R}^{2d}} \phi(x-y)(w-v)f(t,x,v)f(t,y,w)\,\mathrm{d}y\,\mathrm{d}w.$$

- Rigorously justification is called “mean-field limit”.

Relating physical models in different scalings

- From the more macroscopic point of view, the dynamics can be further approximated by fluid equation:

$$\begin{aligned} &\partial_t \rho + \nabla \cdot (\rho u) = 0, \\ &\partial_t (\rho u) + \nabla \cdot (\rho u \otimes u) + \nabla p \\ &\qquad\qquad = \int_{\mathbb{R}^d} \phi(x-y)(u(y)-u(x))\rho(x)\rho(y)\, \mathrm{d}y. \end{aligned}$$

- Obtaining hydrodynamic models from the other model with different scaling (usually, from the kinetic equations) is called "hydrodynamic limit".

Relating physical models in different scalings

- Mean-field limits
  - Finite-time mean-field limit (Ha-Liu 09)
  - Uniform-in-time mean-field limit (Ha-Kim-Zhang 18)
  - Stochastic mean-field limit (Bolley-Canizo-Carrillo 11)
  - Sharp sensitivity regions (Carrillo-Choi-Hauray-Salem 18)
  - ...
- Hydrodynamic limits
  - Hydrodynamic limit to isothermal Euler-alignment model (Karper-Mellet-Trivisa 15)
  - Hydrodynamic limit to drift-type equation (Poyato-Soler 16)
  - Kinetic-fluid coupled models (Carrillo-Choi-Karper 16)
  - Pressureless Euler equation (Figalli-Kang 19)
  - ...

Relating physical models in different scalings

- In 2021, I was looking for some research topic, related to quantum mechanics, and find some references for hydrodynamic limits of Schrödinger equations, Schrödinger-Poisson equations, or Klein-Gordon equations.
- Our question was, what can we say about hydrodynamic limit when the quantum mechanical system is interacting with electromagnetic fields?

Preliminary: Semi-classical limit of the Schördinger equations

Schrödinger equation

- The dynamics of classical particle is goverend by the Newton's second law of motion.
- However, when the time and spatial scales are extremely small, the dynamics of it cannot be described by the classical mechanics.
- Instead, the Schrödinger equation describes the dynamics of the particle:

$$\mathrm{i}\hbar\partial_t\psi + \frac{\hbar^2}{2}\Delta\psi - V\psi = 0,$$

where $\psi : \mathbb{R}_+ \times \mathbb{R}^d \to \mathbb{C}$ is called a wave function of the particle, and $V : \mathbb{R}_+ \times \mathbb{R}^d \to \mathbb{R}$ is a potential.

Schrödinger equation

- In the Schrödinger equation, the physical constant $\hbar$ is the (reduced) Planck constant whose value is $1.055 \times 10^{-34} J \cdot s$.
- Therefore, it is natural to ask what happens to the solution ψ to the Schrödinger equation as $\hbar \to 0$.
- This is called a semi-classical limit of the Schrödinger equations.

Madelung transformation

- Another approach to deal with the semi-classical limit is the hydrodynamic formulation.
- The Madelung transformation:

$$\psi(t,x) = \sqrt{\rho(t,x)}\exp\left(\frac{\mathrm{i}S(t,x)}{\hbar}\right),$$

from which we introduce "hydrodynamic variables":

$$\rho(t,x) := |\psi(t,x)|^2, \quad (\rho u)(t,x) := \rho\nabla_x S = \hbar \mathrm{Im}(\overline{\psi}\nabla_x\psi).$$

Madelung transformation

- Multiply the Schrödinger equation by $\overline{\psi}$ and then take the imaginary part to yield

$$\hbar \mathrm{Re}(\overline{\psi}\partial_t\psi) + \frac{\hbar^2}{2}\mathrm{Im}(\overline{\psi}\Delta\psi) = 0,$$

which can be written as

$$\overline{\psi}\partial_t\psi + \psi\overline{\partial_t\psi} + \hbar \mathrm{Im}(\nabla\cdot(\overline{\psi}\nabla\psi)) = 0,$$

or

$$\partial_t(|\psi|^2) + \nabla\cdot(\hbar \mathrm{Im}(\overline{\psi}\nabla\psi)) = 0.$$

- Therefore, the following continuity equation holds:

$$\partial_t\rho + \nabla\cdot(\rho u) = 0.$$

Madelung transformation

- On the other hand, if we take a gradient to the Schrodinger equation and then multiply $\overline{\psi}$, we have

$$\mathrm{i}\hbar\overline{\psi}\partial_t\nabla\psi + \frac{\hbar^2}{2}\overline{\psi}\nabla\Delta\psi - \overline{\psi}\nabla(V\psi) = 0.$$

- Moreover, if we take conjugate to the Schrödinger equation and then multiply $\nabla\psi$, we have

$$-\mathrm{i}\hbar\nabla\psi\partial_t\overline{\psi} + \frac{\hbar^2}{2}\nabla\psi\overline{\Delta\psi} - V\overline{\psi}\nabla\psi = 0.$$

Madelung transformation

- We take a difference between two equations and take a real part to derive

$$\partial_t \hbar \mathrm{Im}(\overline{\psi}\nabla\psi) + \frac{\hbar^2}{2}\mathrm{Re}(\nabla\psi\overline{\Delta\psi} - \overline{\psi}\nabla\Delta\psi) + \nabla V|\psi|^2 = 0.$$

- This equation can be written as

$$\partial_t(\rho u) + \frac{\hbar^2}{4}\nabla\cdot(2(\nabla\psi\otimes\overline{\nabla\psi} + \overline{\nabla\psi}\otimes\nabla\psi) - \nabla^2|\psi|^2) = -\rho\nabla V,$$

or more formally

$$\partial_t(\rho u) + \nabla\cdot(\rho u\otimes u) = -\rho\nabla V + \frac{\hbar^2}{2}\rho\nabla\left(\frac{\Delta\sqrt{\rho}}{\sqrt{\rho}}\right).$$

Quantum hydrodynamic equations

- Therefore, combining two equations, we can derive the quantum hydrodynamic equations from the Schrödinger equation:

$$\partial_t \rho + \nabla \cdot (\rho u) = 0,$$
$$\partial_t(\rho u) + \nabla \cdot (\rho u \otimes u) = -\rho \nabla V + \frac{\hbar^2}{2} \rho \nabla \left(\frac{\Delta \sqrt{\rho}}{\sqrt{\rho}} \right).$$

- Formally, if we take a limit $\hbar = \varepsilon \to 0$, we obtain the classical Euler equation for fluid:

$$\partial_t \rho + \nabla \cdot (\rho u) = 0,$$
$$\partial_t(\rho u) + \nabla \cdot (\rho u \otimes u) = -\rho \nabla V.$$

Example

- Consider a simple initial data on 1D with $V = 0$:

$$\psi_0(x) := \frac{1}{\sqrt[4]{2\pi}} \exp\left(\frac{ix}{\hbar} - \frac{x^2}{4}\right).$$

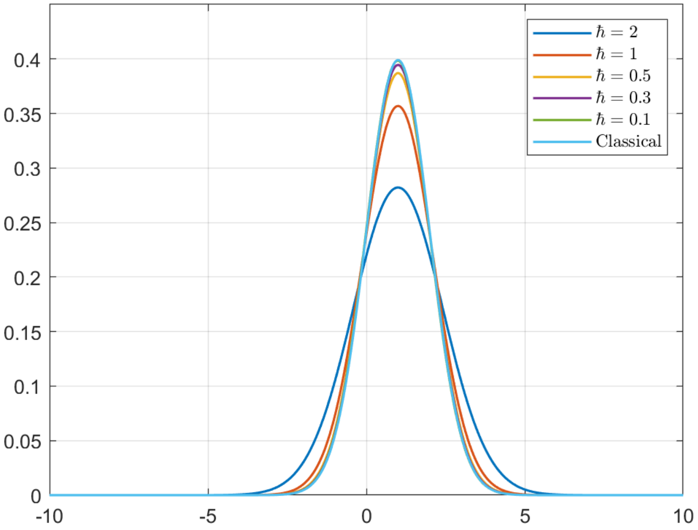
- Then,

$$\rho_0(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right), \quad \rho_0 u_0(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right).$$

- In this case, the velocity $u_0(x) = 1$ is constant and therefore, the solution to the Euler equation is simply given as

$$\rho(t,x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x-t)^2}{2}\right), \quad u(t,x) = 1.$$

Examle



$\hbar = 2$
$\hbar = 1$
$\hbar = 0.5$
$\hbar = 0.3$
$\hbar = 0.1$
Classical
0.4
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
-10
-5
0
5
10

Modulated energy method

- Above, we use the Madelung transformation to derive the quantum hydrodynamic equations. However, this is just a formal derivation since the Madelung transformation does not defined when $\rho = 0$.
- In order to avoid the direct usage of the quantum hydrodynamic formulation, one may use the modulated energy method, that consideres the quantities such as

$$\mathcal{H}^\varepsilon = \frac{\varepsilon^2}{2} \int_{\mathbb{R}^d} |(\nabla - \mathrm{i} u)\psi^\varepsilon|^2 \,\mathrm{d}x + \int_{\mathbb{R}^d} F(|\psi^\varepsilon|^2, \rho) \,\mathrm{d}x.$$

Modulated energy method

- Consider the cubic nonlinear Schrödinger equation:

$$\mathrm{i}\varepsilon\partial_t\psi^\varepsilon + \frac{\varepsilon^2}{2}\Delta\psi^\varepsilon - |\psi^\varepsilon|^2\psi^\varepsilon = 0.$$

- Then, the total energy of the system is given as

$$\mathcal{E}(t) := \frac{\varepsilon^2}{2}\int_{\mathbb{R}^d} |\nabla\psi^\varepsilon|^2\,\mathrm{d}x + \int_{\mathbb{R}^d} |\psi^\varepsilon|^4\,\mathrm{d}x.$$

- Then, the modulated energy, comapred to the limiting hydrodynamic equation for (ρ, u) is given as

$$\mathcal{H}^\varepsilon = \frac{\varepsilon^2}{2}\int_{\mathbb{R}^d} |(\nabla - \mathrm{i}u)\psi^\varepsilon|^2\,\mathrm{d}x + \int_{\mathbb{R}^d} (|\psi^\varepsilon|^2 - \rho)^2\,\mathrm{d}x.$$

Modulated energy method

- In terms of the hydrodynamic variables, we have

$$\mathcal{H}^\varepsilon = \frac{1}{2}\int_{\mathbb{R}^d} \rho^\varepsilon |u^\varepsilon - u|^2 \, dx + \int_{\mathbb{R}^d} (\rho^\varepsilon - \rho)^2 \, dx + \frac{\varepsilon^2}{2}\int_{\mathbb{R}^d} |\nabla\sqrt{\rho^\varepsilon}|^2 \, dx.$$

- This provide a pseudo-metric between the hydrodynamic variables $(\rho^\varepsilon, u^\varepsilon)$ and (ρ, u). Indeed, if $\mathcal{H}^\varepsilon$ vanishes as $\varepsilon \to 0$, one can show that $(\rho^\varepsilon, \rho^\varepsilon u^\varepsilon)$ converges to $(\rho, \rho u)$ in an appropriate way.
- Therefore, the main goal of the modulated energy method is to estimate the quantity such as $\mathcal{H}^\varepsilon$.

Hydrodynamic limit of the Maxwell-Klein-Gordon equations

Klein-Gordon equations

- The Klein-Gordon equation that describes the dynamics of the relativistic quantum particle reads as

$$\frac{\hbar^2}{2}\partial_\mu\partial^\mu\phi + \frac{c^2}{2}\phi + V'(|\phi|^2)\phi = 0,$$

where ϕ is a matter field, c is the speed of light and $x_0 = ct$.

- Under the interaction with the electromagnetic field, the governing equation for the particle is chagned by the following minimal coupling rule:

$$\partial_\mu \to \partial_\mu - \frac{\mathrm{i}}{c\hbar}A_\mu, \quad \mu = 0, 1, 2, 3.$$

- $A_\mu = (A_0, A)$ is the electromagnetic potential, called gauge.

Maxwell-Klein-Gordon equations

- The Maxwell-Klein-Gordon (MKG) equations:

$$\begin{aligned}
&\frac{\hbar^2}{2} D_\mu D^\mu \phi + \frac{c^2}{2}\phi + V'(|\phi|^2)\phi = 0, \quad (t,x) \in \mathbb{R}_+ \times \mathbb{R}^3, \\
&\frac{1}{c}\partial_t(\nabla \cdot A) - \Delta A_0 = \frac{\hbar}{c}\mathrm{Im}(\overline{\phi} D_0 \phi), \\
&\left(\frac{1}{c^2}\partial_t^2 A - \Delta A\right) - \nabla\left(\frac{1}{c}\partial_t A_0 - \nabla \cdot A\right) = \frac{\hbar}{c}\mathrm{Im}(\overline{\phi} D \phi).
\end{aligned}$$

- Here, the covariant derivative is $D_\mu = \partial_\mu - \frac{\mathrm{i}}{c\hbar}A_\mu$, and the nonlinear potential $V(\rho) = \frac{1}{\gamma-1}\rho^\gamma$.

Gauge invariance

- The MKG equation is invariant under the following transformation:

$$\phi \to \phi e^{-\mathrm{i}\chi}, \quad A_\mu \to A_\mu + \hbar \partial_\mu \chi,$$

  where χ is an arbitrary smooth function.
- Therefore, we need to fix the gauge via extra gauge condition. Usually, the following two conditions are considered:
  1. The Coulomb gauge condition: $\nabla \cdot A = 0$,
  2. The Lorenz gauge condition: $\partial_t A_0 - \nabla \cdot A = 0$.
- We only consider the case of Coulomb gauge condition.

Previous results

- Nonrelativistic limit ($c \to +\infty$)
  - Klein-Gordon equation to Schrödinger equation (Schoene 79)
  - Maxwell-Klein-Gordon equation to Schrödinger-Poisson equation (Bechouche-Mauser-Selberg 04, Masmoudi-Nakanishi 03)
- Semi-classical limit ($\hbar \to 0$)
  - Schrödinger-Poisson to compressible Euler equations (Jüngel-Wang 03)
  - Schrödinger-Poisson to incompressible Euler equations (Puel 02)
  - Schrödinger-Poisson to Euler-Poisson (Li-Lin 03)
- Simultaneous limit ($c \to +\infty, \hbar \to 0$)
  - Klein-Gordon to compressible/incompressible Euler equations (Lin-Wu 12)

Modulated wave function

- Introducing the modulated wave function

$$\psi(t,x) = \phi(t,x)\exp\left(\frac{ic^2 t}{\hbar}\right),$$

the MKG system under the Coulomb gauge condition $\nabla \cdot A = 0$ can be written as

$$\begin{aligned}
&ic\hbar D_0\psi - \frac{\hbar^2}{2}D_0^2\psi + \frac{\hbar^2}{2}D^2\psi - V'(|\psi|^2)\psi = 0,\\
&-\Delta A_0 = -|\psi|^2 + \frac{\hbar}{c}\mathrm{Im}(\overline{\psi}D_0\psi),\\
&\frac{1}{c^2}\partial_t^2 A - \Delta A - \frac{1}{c}\partial_t \nabla A_0 = \frac{\hbar}{c}\mathrm{Im}(\overline{\psi}D\psi).
\end{aligned}$$

Well-posedness of the MKG equations

- In order to pass the limit $\varepsilon \to 0$, we first need to guarantee the existence of the nonlinear MKG equation.

Theorem

Let $1 < \gamma < 3$. Suppose that the initial data $(\phi(0,\cdot), A(0,\cdot), \partial_t \phi(0,\cdot), \partial_t A(0,\cdot))$ satisfy

$$\|\phi(0,\cdot)\|_{H^2} + \|\partial_t \phi\|_{H^1} + \|\nabla A(0,\cdot)\|_{H^1} + \|\partial_t A(0,\cdot)\|_{H^1} < +\infty.$$

Then, for any $T > 0$, there exists a unique classical solution (ϕ, A_α) to the MKG equation.

Scaled electrostatic Maxwell-Klein-Gordon equations

- Below, we consider the electrostatic case, that is $A \equiv 0$.
- Consider the scales $\hbar = \varepsilon$ and $c = \varepsilon^{-\kappa}$ with $\kappa > 0$, the MKG system can be written as

$$
\begin{aligned}
&\mathrm{i}\varepsilon^{1-\kappa} D_0^\varepsilon \psi^\varepsilon - \frac{\varepsilon^2}{2}(D_0^\varepsilon)^2 \psi^\varepsilon + \frac{\varepsilon^2}{2}\Delta\psi^\varepsilon - V'(|\psi^\varepsilon|^2)\psi^\varepsilon = 0, \\
&-\Delta A_0^\varepsilon = -|\psi^\varepsilon|^2 + \varepsilon^{1+\kappa}\mathrm{Im}(\overline{\psi^\varepsilon} D_0^\varepsilon \psi^\varepsilon),
\end{aligned}
$$

  where $D_0^\varepsilon := \partial_0 - \frac{\mathrm{i}}{\varepsilon^{1-\kappa}} A_0^\varepsilon$.
- This implies that we are considering the case when the Planck constant and the speed of light tends to 0 and infinity simultaneously. (Polynomial relation is not important!)

Conservation laws

- We define hydrodynamic quantities

$$\rho^\varepsilon := |\psi^\varepsilon|^2, \quad J^\varepsilon = \rho^\varepsilon u^\varepsilon := \varepsilon \mathrm{Im}(\overline{\psi^\varepsilon} \nabla \psi^\varepsilon),$$

and their relativistic corrections:

$$\rho_R^\varepsilon := \varepsilon^{1+\kappa} \mathrm{Im}(\overline{\psi^\varepsilon} D_0^\varepsilon \psi^\varepsilon), \quad J_R^\varepsilon := \varepsilon^{2+\kappa} \mathrm{Re}(D_0^\varepsilon \psi^\varepsilon \overline{\nabla \psi^\varepsilon}).$$

- Also, we define electric field $E^\varepsilon := \nabla A_0^\varepsilon$.
- Then, we have the conservation of mass and energy:

$$\frac{\mathrm{d}}{\mathrm{d}t} \int_{\mathbb{R}^3} (\rho^\varepsilon - \rho_R^\varepsilon)\,\mathrm{d}x = 0,$$
$$\frac{\mathrm{d}}{\mathrm{d}t} \int_{\mathbb{R}^3} \left(\frac{\varepsilon^2}{2} (D_0^\varepsilon \psi^\varepsilon)^2 + \frac{\varepsilon^2}{2} |\nabla \psi^\varepsilon|^2 + \frac{1}{2} |\nabla A_0^\varepsilon|^2 + V(|\psi^\varepsilon|^2) \right) \mathrm{d}x = 0.$$

Hydrodyanamic formulation

- After tedious verification, we obtain (at least formally), the hydrodynamic equations for the MKG system:

$$
\begin{aligned}
&\partial_t(\rho^\varepsilon - \rho_R^\varepsilon) + \nabla\cdot(\rho^\varepsilon u^\varepsilon) = 0,\\
&\partial_t((\rho^\varepsilon - \rho_R^\varepsilon)u^\varepsilon) + \nabla\cdot(\rho^\varepsilon u^\varepsilon \otimes u^\varepsilon) + \nabla p(\rho^\varepsilon)\\
&\qquad\qquad = (\rho^\varepsilon - \rho_R^\varepsilon)\nabla A_0^\varepsilon + \frac{\rho^\varepsilon \varepsilon^2}{2}\nabla\left(\frac{\Box_\kappa\sqrt{\rho^\varepsilon}}{\sqrt{\rho^\varepsilon}}\right),\\
&\Delta A_0^\varepsilon = \rho^\varepsilon - \rho_R^\varepsilon.
\end{aligned}
$$

- This formally converges to the Euler-Poisson system:

$$
\begin{aligned}
&\partial_t\rho + \nabla\cdot(\rho u) = 0,\\
&\partial_t(\rho u) + \nabla\cdot(\rho u\otimes u) + \nabla p(\rho) = \rho\nabla A_0,\\
&\Delta A_0 = \rho.
\end{aligned}
$$

Main theorem

- We consider the modulated energy $\mathcal{H}^\varepsilon$ defined as

$$\mathcal{H}^\varepsilon(t) := \int_{\mathbb{R}^3} \frac{|(\varepsilon\nabla - \mathrm{i}u)\psi^\varepsilon|^2}{2}\,\mathrm{d}x + \int_{\mathbb{R}^3} \frac{|\varepsilon D_0^\varepsilon \psi^\varepsilon|^2}{2}\,\mathrm{d}x + \int_{\mathbb{R}^3} \frac{p(\rho^\varepsilon|\rho)}{\gamma - 1}\,\mathrm{d}x + \int_{\mathbb{R}^3} \frac{|\nabla A_0^\varepsilon - \nabla A_0|^2}{2}\,\mathrm{d}x.$$

Theorem

Suppose $1 < \gamma < 3$. Let $(\psi^\varepsilon, A_0^\varepsilon)$ be the global solution to the (electrostatic) MKG equations. Moreover, let (ρ, u, A_0) be the unique local-in-time smooth solution to the Euler-Poisson equations for $0 \le t \le T_*$. Suppose that the initial data are well-prepared in the sense that $\mathcal{H}^\varepsilon(0) \le C\varepsilon^\lambda$.

Main theorem

Theorem (continued)

Then, for any $0 \leq t \leq T_*$, we have

$$
\begin{aligned}
&\rho^\varepsilon(t,\cdot) \to \rho(t,\cdot), \quad \text{in} \quad L^\gamma(\mathbb{R}^3),\\
&(\rho^\varepsilon u^\varepsilon)(t,\cdot) \to (\rho u)(t,\cdot), \quad \text{in} \quad L^{\frac{2\gamma}{\gamma+1}}(\mathbb{R}^3),\\
&(\sqrt{\rho^\varepsilon} u^\varepsilon)(t,\cdot) \to (\sqrt{\rho} u)(t,\cdot), \quad \text{in} \quad L^2(\mathbb{R}^3),\\
&\nabla A_0^\varepsilon \to \nabla A_0, \quad \text{in} \quad L^2(\mathbb{R}^3),
\end{aligned}
$$

as $\varepsilon \to 0$, if $2 \leq \gamma < 3$ and the case of $1 < \gamma < 2$ is the same, with the local-in-space convergence.

Modulated energy

- Recall that the modulated energy for the MKG system is given as

$$\mathcal{H}^\varepsilon(t) = \frac{1}{2}\int_{\mathbb{R}^3} |(\varepsilon\nabla - \mathrm{i}u)\psi^\varepsilon|^2\,\mathrm{d}x + \frac{1}{2}\int_{\mathbb{R}^3} |\varepsilon D_0^\varepsilon \psi^\varepsilon|^2\,\mathrm{d}x + \int_{\mathbb{R}^3} \frac{p(\rho^\varepsilon|\rho)}{\gamma - 1}\,\mathrm{d}x + \frac{1}{2}\int_{\mathbb{R}^3} |\nabla A_0^\varepsilon - \nabla A_0|^2\,\mathrm{d}x.$$

- Now, we directly estimate the modulated energy, instead of using the hydrodynamic quantitiy.
- We need the following auxiliary term $\mathcal{G}^\varepsilon$ defined as

$$\mathcal{G}^\varepsilon(t) = \int_{\mathbb{R}^3} u\cdot J_R^\varepsilon\,\mathrm{d}x - \frac{1}{2}\int_{\mathbb{R}^3} \rho_R^\varepsilon|u|^2\,\mathrm{d}x + \frac{\varepsilon^{2+2\kappa}}{4}\int_{\mathbb{R}^3} \partial_t\rho^\varepsilon\nabla\cdot u\,\mathrm{d}x + \frac{\gamma}{\gamma-1}\int_{\mathbb{R}^3} \rho^{\gamma-1}\rho_R^\varepsilon\,\mathrm{d}x$$

Modulated energy estimate

Proposition

The quantities $\mathcal{H}^\varepsilon$ and $\mathcal{G}^\varepsilon$ satisfy

$$\frac{\mathrm{d}}{\mathrm{d}t}(\mathcal{H}^\varepsilon + \mathcal{G}^\varepsilon) \leq C\mathcal{H}^\varepsilon + C\varepsilon^{\min\{1,\kappa\}}.$$

- One can show that $\mathcal{G}^\varepsilon$ can be bounded by ε^κ, which implies that

$$\begin{aligned}\mathcal{H}^\varepsilon(t) &\leq \mathcal{H}^\varepsilon(0) - (\mathcal{G}^\varepsilon(t) - \mathcal{G}^\varepsilon(0)) + C\int_0^t \mathcal{H}^\varepsilon(s)\,\mathrm{d}s + C\varepsilon^{\min\{1,\kappa\}} \\ &\leq C\varepsilon^{\min\{1,\kappa,\lambda\}} + C\int_0^t \mathcal{H}^\varepsilon(s)\,\mathrm{d}s.\end{aligned}$$

- Then Grönwall's inequality implies $\mathcal{H}^\varepsilon \to 0$.

Proof of Theorem

- Convergence of the density: if $\gamma \geq 2$, we have

$$\|\rho^\varepsilon - \rho\|_{L^\gamma}^\gamma \leq C \int_{\mathbb{R}^3} p(\rho^\varepsilon|\rho)\,\mathrm{d}x \leq \mathcal{H}^\varepsilon \to 0, \quad \text{as} \quad \varepsilon \to 0.$$

- Convergence of the momentum:

$$\begin{aligned} &\|\rho^\varepsilon u^\varepsilon - \rho u\|_{L^{\frac{2\gamma}{\gamma+1}}} \\ &\leq \|\rho^\varepsilon (u^\varepsilon - u)\|_{L^{\frac{2\gamma}{\gamma+1}}} + \|(\rho^\varepsilon - \rho) u\|_{L^{\frac{2\gamma}{\gamma+1}}} \\ &\leq \|\sqrt{\rho^\varepsilon}\|_{L^{2\gamma}} \|\sqrt{\rho^\varepsilon}|u^\varepsilon - u|\|_{L^2} + \|\rho^\varepsilon - \rho\|_{L^\gamma} \|u\|_{L^{\frac{2\gamma}{\gamma-1}}} \\ &\leq C\|\sqrt{\rho^\varepsilon}|u^\varepsilon - u|\|_{L^2} + C\|\rho^\varepsilon - \rho\|_{L^\gamma} \leq C\mathcal{H}^\varepsilon \to 0. \end{aligned}$$

Proof of Theorem

- The convergence of electromagnetic fields are directly comes from the modulated energy estimate.
- The convergence of the relativistic hydrodynamic quantities can be shown as

$$\int_{\mathbb{R}^3} |J_R^\varepsilon| \, \mathrm{d}x \le \varepsilon^\kappa \|\varepsilon D_0^\varepsilon \psi^\varepsilon\|_{L^2} \|\varepsilon D^\varepsilon \psi^\varepsilon\|_{L^2} \le C\varepsilon^\kappa,$$

and

$$\int_{\mathbb{R}^3} |\rho_R^\varepsilon|^p \, \mathrm{d}x = \int_{\mathbb{R}^3} \varepsilon^{\kappa p} |\psi^\varepsilon|^p |\varepsilon D_0^\varepsilon \psi^\varepsilon|^p \, \mathrm{d}x \le C\varepsilon^{\kappa p} \|\rho^\varepsilon\|_{L^{\frac{p}{2-p}}}^{p/2}.$$

We choose $p = \frac{2\gamma}{\gamma+1}$ to obtain $\|\rho_R^\varepsilon\|_{L^{\frac{2\gamma}{\gamma+1}}} \le C\varepsilon^\kappa$.

- This completes the proof of hydrodynamic limits.

Thank you very much for your attention

and

Happy 60th birthday, Prof. Yu!